Brief / Requirements

Intended use: requirements baseline

01 / What is going wrong?

The phrase 12 MW means different things to the owner and design team.

02 / The controlled response

Confirm the IT-load definition, phase boundary and acceptance measures before design proceeds.

03 / Who must provide what?

1. OWN - Owner-approved OPR with defined load terminology and assumptions

Evidence register: DC01-OWN-SITE-ZZ-RP-O-0002.pdf | R02 | simulated receipt, not a real inspection result.

2. OPS - Operator review with measurable acceptance requirements

Evidence register: DC01-OPS-SITE-ZZ-RP-O-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OWN. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

TRAINING ONLY / UNSIGNED / NO CONSTRUCTION OR TEST AUTHORITY

Project codes and responsibilities are defined only for this fictional case.

Concept Design

Intended use: concept selection

01 / What is going wrong?

A and B paths appear separate on the single-line diagram, but occupy a common spatial risk area.

02 / The controlled response

Review common-mode assumptions across electrical layout and fire compartments; obtain the named risk decision.

03 / Who must provide what?

1. ELD - Topology and spatial-route option review

Evidence register: DC01-ELD-E01-00-RP-E-0001.pdf | R02 | simulated receipt, not a real inspection result.

2. FLS - Fire interface and recorded residual-risk disposition

Evidence register: DC01-FLS-E01-00-RP-F-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OWN. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Schematic Design

Intended use: schematic coordination

01 / What is going wrong?

Models appear aligned in a federation, while discipline exports use different coordinate bases.

02 / The controlled response

Establish one documented survey/control reference and verify an actual exchange against it.

03 / Who must provide what?

1. IMT - Coordinate and level register with source versions

Evidence register: DC01-IMT-DH1-00-RP-X-0001.pdf | R02 | simulated receipt, not a real inspection result.

2. SUR - Survey-based exchange validation and exceptions

Evidence register: DC01-SUR-DH1-00-RP-V-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: PMO. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

Design Development

Intended use: design development review

01 / What is going wrong?

A new single-line diagram is paired with an old load schedule and plan.

02 / The controlled response

Reconcile the dependency set and issue a package manifest with exact revisions and intended use.

03 / Who must provide what?

1. ELD - SLD, load schedule and model consistency review

Evidence register: DC01-ELD-E01-00-RP-E-0002.pdf | R02 | simulated receipt, not a real inspection result.

2. IMT - Transmittal manifest identifying exact representations and revisions

Evidence register: DC01-IMT-E01-00-RP-X-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: PMO. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

Tender / Bidding

Intended use: bidding only

01 / What is going wrong?

Bidders received different addenda and therefore priced different scope.

02 / The controlled response

Use controlled addendum distribution and acknowledged baseline reconciliation before comparison.

03 / Who must provide what?

1. PMO - Addendum distribution and recipient acknowledgements

Evidence register: DC01-PMO-SITE-ZZ-RP-X-0002.pdf | R02 | simulated receipt, not a real inspection result.

2. QSX - Reconciled tender pricing baseline and exceptions

Evidence register: DC01-QSX-SITE-ZZ-RP-Q-0002.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OWN. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

Permit / Authority Review

Intended use: permit submission

01 / What is going wrong?

An authority response covers one submission scope, but the team treats it as blanket permission.

02 / The controlled response

Track the scope and conditions of the simulated response and assess affected packages separately.

03 / Who must provide what?

1. PMO - Authority scope and conditions register

Evidence register: DC01-PMO-SITE-ZZ-RP-X-0003.pdf | R02 | simulated receipt, not a real inspection result.

2. ARC - Affected-package hold/release decision with named responsibility

Evidence register: DC01-ARC-SITE-ZZ-RP-A-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: REG. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

Procurement / Vendor Design

Intended use: procurement / fabrication release

01 / What is going wrong?

A substitute UPS has the same headline rating but different clearances and interfaces.

02 / The controlled response

Assess the deviation against electrical, cooling, architecture and maintenance interfaces before acceptance.

03 / Who must provide what?

1. VUP - Certified substitute equipment interface package

Evidence register: DC01-VUP-E01-00-RP-E-0002.pdf | R02 | simulated receipt, not a real inspection result.

2. ELD - Multidiscipline deviation decision and limitations

Evidence register: DC01-ELD-E01-00-RP-E-0004.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: GCC. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Shop Drawing / Coordination

Intended use: installation package review

01 / What is going wrong?

A clash is removed by moving busway into a common segment of A and B routes.

02 / The controlled response

Recheck design intent and segregation assumptions; clash-free geometry alone is insufficient.

03 / Who must provide what?

1. ELS - Busway-route and design-intent review

Evidence register: DC01-ELS-DH1-00-RP-E-0001.pdf | R02 | simulated receipt, not a real inspection result.

2. ELD - Accepted deviation with affected-system evidence

Evidence register: DC01-ELD-DH1-00-RP-E-0002.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: GCC. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Construction / Site Control

Intended use: construction for defined scope

01 / What is going wrong?

The CDE has withdrawn an old issue, but a foreman still uses the PDF received in a chat.

02 / The controlled response

Control the affected workfront, recall the obsolete copy and verify recipients changed revision; inspect affected work.

03 / Who must provide what?

1. GCC - Distribution and recall acknowledgements

Evidence register: DC01-GCC-E01-00-RP-X-0001.pdf | R02 | simulated receipt, not a real inspection result.

2. ELS - Affected-work inspection and corrective-action evidence

Evidence register: DC01-ELS-E01-00-RP-E-0002.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: GCC. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Commissioning / Integrated Tests

Intended use: commissioning evidence acceptance

01 / What is going wrong?

Equipment tests passed, but an integrated scenario retains an unresolved exception.

02 / The controlled response

Separate test coverage and configuration; obtain a scoped decision on residual exceptions and retest needs.

03 / Who must provide what?

1. CXP - Integrated-test coverage and traceable results

Evidence register: DC01-CXP-DH1-00-RP-X-0003.pdf | R02 | simulated receipt, not a real inspection result.

2. BMS - Exception disposition and retest acceptance

Evidence register: DC01-BMS-DH1-00-RP-B-0002.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OWN. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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As-Built / Verified Record

Intended use: verified record information

01 / What is going wrong?

A bypass valve installed under an accepted field change is absent from the record model.

02 / The controlled response

Reconcile the field-change register with installed evidence and record verification coverage and limitations.

03 / Who must provide what?

1. MCS - Field-change to record-model reconciliation

Evidence register: DC01-MCS-CHY-00-RP-M-0005.pdf | R02 | simulated receipt, not a real inspection result.

2. SUR - Survey or inspection verification with coverage statement

Evidence register: DC01-SUR-CHY-00-RP-V-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: GCC. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Handover / Asset Acceptance

Intended use: handover acceptance

01 / What is going wrong?

Manuals are present, but the operator cannot find the applicable manual from an AssetID.

02 / The controlled response

Test the actual find-and-use path with the recipient, including model, serial, warranty and location applicability.

03 / Who must provide what?

1. GCC - Asset-to-document linkage and applicability

Evidence register: DC01-GCC-E01-00-RP-X-0002.pdf | R02 | simulated receipt, not a real inspection result.

2. OPS - Operator find-and-use acceptance evidence

Evidence register: DC01-OPS-E01-00-RP-O-0004.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OPS. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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FM / Operational Change

Intended use: operational use / change control

01 / What is going wrong?

A replacement component is recorded in email but not in asset or warranty records.

02 / The controlled response

Keep persistent asset identity and track the component/serial change with effective dates and applicability.

03 / Who must provide what?

1. OPS - Service and component replacement record

Evidence register: DC01-OPS-E01-00-RP-O-0006.pdf | R02 | simulated receipt, not a real inspection result.

2. VUP - CMMS and warranty reconciliation

Evidence register: DC01-VUP-E01-00-RP-E-0003.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OPS. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

What prevents bypass? Who will detect the wrong copy, identity or configuration? Who will hold, notify, correct and verify closure?

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Digital Twin / Data Trust

Intended use: decision-support data acceptance

01 / What is going wrong?

The viewer displays UPS-A-01, but its value originates from UPS-B-01.

02 / The controlled response

Validate the canonical identity chain from asset to model, CMMS and source point before decision support.

03 / Who must provide what?

1. DTI - AssetID, IFC GUID and point-mapping validation

Evidence register: DC01-DTI-E01-00-RP-X-0001.pdf | R02 | simulated receipt, not a real inspection result.

2. BMS - Source-owner identity reconciliation and limitation record

Evidence register: DC01-BMS-E01-00-RP-B-0001.pdf | R02 | simulated receipt, not a real inspection result.

04 / The acceptance boundary

Named receiving authority in this case: OPS. Accept only this information-control issue, within the stated use and recorded baseline. File upload, a folder move or technical administration does not establish that authority.

Revision control: changing a referenced source or evidence revision makes the previous acceptance stale. Review the affected scope and obtain a new decision; do not overwrite the historical record.

Discuss with the team

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Assumptions & sources

12 MW is illustrative IT load: 2 halls x 500 racks x 12 kW. It is not facility demand, a sizing calculation or a certified resilience rating.

Permit, tender, procurement and phased construction can overlap. The case is not a mandatory sequential project schedule.

The IFC contains 24 proxy boxes with fictional asset properties. It is not the geometry of the native models listed in the register. The BCF camera/snapshot is schematic.

JSON samples are synthetic and read-only. No real credentials, OT endpoints, control commands, tenant data or site location are included.

Autodesk Docs - naming template and related attributes

https://help.autodesk.com/cloudhelp/ENU/Docs-Files/files/files-naming-standard/Set_Up_Naming_Standard.html

WBDG - Commissioning documents

https://legacy.wbdg.org/building-commissioning/commissioning-documents

buildingSMART - BCF XML

https://github.com/buildingSMART/BCF-XML

NIBS - COBie V3

https://nbims.nibs-wbdg.org/nbims/v3/cobie/

NIST - Digital twins

https://www.nist.gov/digital-twins

NIST IR 8356 - security and trust considerations

https://www.nist.gov/news-events/news/2025/02/security-and-trust-considerations-digital-twin-technology-nist-releases-ir